

```

def showInfo(self): ❹
    print()
    print('TV Status:')
    if self.isOn:
        print('    TV is: On')
        print('    Channel is:', self.channelList[self.channelIndex])
        if self.isMuted:
            print('    Volume is:', self.volume, '(sound is muted)')
        else:
            print('    Volume is:', self.volume)
    else:
        print('    TV is: Off')

```

Listing 2-7: The `TV` class with many instance variables and methods

The `__init__()` method ❶ creates all the instance variables used in all the methods and assigns reasonable starting values to each. Technically, you can create an instance variable inside any method; however, it is a good programming practice to define all instance variables in the `__init__()` method. This avoids the risk of an error when attempting to use an instance variable in a method before it's been defined.

The `power()` method ❷ represents what happens when you push the power button on a remote. If the TV is off, pushing the power button turns it on; if the TV is on, pushing the power button turns it off. To code this behavior I've used a *toggle*, which is a Boolean that's used to represent one of two states and can easily be switched between them. With this toggle, the `not` operator switches the `self.isOn` variable from `True` to `False`, or from `False` to `True`. The `mute()` method code ❸ does a similar thing, with the `self.muted` variable toggling between muted and not-muted, but first has to check that the TV is on. If the TV is off, calling the `mute()` method has no effect.

One interesting thing to note is that we don't really keep track of the current channel. Instead, we keep track of the *index* of the current channel, which allows us to get the current channel at any time by using `self.channelList[self.channelIndex]`.

The `channelUp()` ❹ and `channelDown()` ❺ methods basically increment and decrement the channel index, but there is also some clever code in them to allow for wrap-around. If you're currently at the last index in the channel list and the user asks to go to the next channel up, the TV goes to the first